Field Hours

CSCE 671 Computer-Human Interaction

Original notes for this site. Same topics as the catalog course. Not a reprint of another university's lectures, and not academic credit.

Lecture 1 · 90 minutes

Three ideas that are still questions

Keep the idea from each historical system, not the hardware list. In 1945 Bush described the Memex as trails that link documents, an associative path rather than a pile of files. In 1963 Sutherland's Sketchpad let a person edit a drawing while constraints updated related geometry. One bets on links between records. The other bets on a picture that stays consistent as you drag it. You should know which bet your screen inherited.

Engelbart's 1968 demo put a mouse, a chorded keyset, and a shared view in front of a group. Keep the claim about augmentation, not the novelty of the mouse. A tool can make a new task possible and make an old task harder at the same time. You ask both questions. If you remember only the pointing device, you have stored a product fact and lost the 1968 argument.

Three dates, three claims.

1945: trails between documents

1963: a picture under constraints

1968: a group, not only a mouse

Ask what task became possible

Which bet the screen inherited

1. A document reader stores associative trails between notes. That is the 1945 bet, links between records.

2. A drawing tool updates a second shape when you drag the first, because a constraint still holds. That is the 1963 bet.

3. A shared cursor in a meeting is closer to the 1968 claim, augmentation of a group. Remembering only the mouse drops that claim.

Problem 1. What idea do you keep from the 1945 Memex, and what idea do you keep from the 1963 Sketchpad, if you refuse a gadget list?

Answer. From 1945, trails that link documents, an associative path rather than a pile of files. From 1963, a drawing you can edit while constraints keep related geometry consistent. They are different bets, and a screen can inherit either.

Problem 2. Why is remembering only the mouse a bad summary of the 1968 demo?

Answer. The demo put a shared view in front of a group. The claim is augmentation: a tool can make a new task possible and an old task harder. The mouse is a product fact. You ask both questions, or you

have lost the argument.

Lecture 2 · 90 minutes

Three processors and a bound

The Model Human Processor is a rough estimate, not a portrait. It gives you a perceptual processor, a cognitive processor, and a motor processor. Common cycle times used for this estimate are 100 milliseconds, 70 milliseconds, and 70 milliseconds. A simple reaction, see a light and press a known key, spends about one cycle of each. The sum is 240 milliseconds. You use it to spot a demand for 3 decisions inside 100 milliseconds.

Do not add this estimate to a later keystroke model as if they timed the same event twice. The 240 millisecond figure is one reaction to one signal. A keystroke estimate uses its own operators, including about 200 milliseconds for a key, and it answers a different question. Adding them double-counts the motor act. Use the processor model when the budget is a few tenths of a second. Use keystrokes when you compare 2 methods that take seconds.

One cycle of each processor.

1. Perceive the light, 100 ms

2. Decide it is the signal, 70 ms

3. Press the key, 70 ms

4. Sum to 240 ms

A budget of 100 milliseconds

1. A simple reaction is about 100 + 70 + 70 = 240 milliseconds.

2. The game asks for 3 decisions inside 100 milliseconds. Even one reaction does not fit in 100.

3. You call the demand impossible under this estimate. You do not use the estimate to rank two real people.

Problem 1. Using 100, 70, and 70 milliseconds, what is the estimate for one simple reaction, and why is 3 decisions inside 100 milliseconds not credible here?

Answer. 240 milliseconds, the sum of one perceptual, one cognitive, and one motor cycle. Three decisions inside 100 milliseconds is below even one of those reactions. The figure spots an impossible demand. It does not rank two real users.

Problem 2. Why do you not add the 240 millisecond reaction to a 200 millisecond keystroke operator for the same press?

Answer. They answer different questions and the motor act would be counted twice. Use the processor model for a budget of a few tenths of a second. Use keystrokes when you compare 2 methods that take seconds.

Lecture 3 · 90 minutes

Execution and evaluation

Separate the path from goal to action and the path from system state back to the goal. The gulf of execution is the gap between what you want and the action you can find. The gulf of evaluation is the gap between a change and your ability to tell that it matched the goal. A labeled save control can be 1 click, a small execution gulf. You can perform the click and still not know whether the file was stored.

Bridging execution means making the next action discoverable and in the user's terms. Bridging evaluation means showing state in the user's terms, soon enough to act on. A spinner that never ends fails evaluation even if the command was easy to give. A status line that says 'error 17' may fail evaluation if the user cannot map 17 onto a repair. You ask, for each step, which gulf is larger. The 1 click was never the whole complaint.

Two gaps on one save.

Execution

Find the action

Here it is 1 click

The label is clear

Evaluation

See the result

No confirmation

You cannot tell

One click, no confirmation

1. The goal is to store the file. The control is labeled and takes 1 click. The execution gulf is small.

2. Nothing on screen changes to say the write finished. The evaluation gulf is the remaining problem.

3. You can perform the click and still not know whether the file was stored. Polishing the click does not close that gulf.

Problem 1. Save is 1 labeled click, and the screen never confirms the write. Which gulf is larger, and why can the click still fail the user?

Answer. The evaluation gulf is larger. Execution is 1 discoverable click. You can perform it and still not know whether the file was stored, because nothing maps the change back to the goal.

Problem 2. A status line says 'error 17' after an easy command. Which gulf fails, and what would bridge it?

Answer. Evaluation. The user cannot map 17 onto a repair, so the state is not in the user's terms. Bridging evaluation means showing the state in those terms, soon enough to act on. The easy command was the execution side.

Lecture 4 · 90 minutes

Operators you can add

GOMS names goals, operators, methods, and selection rules. The keystroke-level model adds up operators. Use K = 0.2 seconds for a key or button, P = 1.1 seconds to point, H = 0.4 seconds to home the hand, and M = 1.35 seconds for one mental chunk. You put one M on a decision, not on every key inside the chunk. Write the operator sequence before you add. A method you cannot write down cannot be timed.

Compare two methods for one goal, with the hand already on the right device. A recalled command of 4 letters plus Enter is M + 5K = 1.35 + 1.0 = 2.35 seconds. A two-level menu, point and click twice, is M + P + K + P + K = 3.95 seconds. The command wins by 1.6 seconds in the model. You only get that win if the user recalls the 4 letters.

One goal, two timed methods.

Recalled command

M + 5K

2.35 seconds

Needs recall

Two-level menu

Two points and two clicks

3.95 seconds

Needs recognition

Command against a menu

1. Hand on the keyboard. One mental chunk, then 4 letters and Enter: M + 5K = 1.35 + 5 times 0.2 = 2.35 seconds.

2. Hand on the mouse. One mental chunk, two points, two clicks: 1.35 + 1.1 + 0.2 + 1.1 + 0.2 = 3.95 seconds.

3. The difference is 3.95 - 2.35 = 1.6 seconds, and only while the 4 letters are recalled. Looking them up needs more operators.

Problem 1. With K = 0.2 and M = 1.35, the hand already on the keyboard, what is M + 5K for a 4-letter command plus Enter?

Answer. 2.35 seconds. Five keys are 5 times 0.2 = 1.0, and 1.35 + 1.0 = 2.35. You place one M on the chunk, not one M on each key.

Problem 2. The menu method is M + P + K + P + K with P = 1.1. How long is it, and by how much does the recalled command beat it?

Answer. 3.95 seconds: 1.35 + 1.1 + 0.2 + 1.1 + 0.2. The command is 2.35, so it wins by 1.6 seconds in the model, and only if the user recalls the 4 letters. Looking the command up is not charged.

Lecture 5 · 90 minutes

Subtasks and a plan

Hierarchical task analysis splits a goal into subtasks plus a plan. The plan states order, optional steps, and branches. Take 4 subtasks for tea: fill, boil, bag in cup, pour. Fill and bag-in-cup can swap. Pour cannot precede boil, and it cannot precede the bag. You write those constraints. A flat list of the 4 names hides which swaps are legal. The plan is the analysis.

Plans include error paths. If the kettle is empty when you pour, return to fill. Number the steps and write the rule in one line: do 1 through 4, but if water is missing redo 1 before 4. A colleague can test it by asking whether step 4 is legal when step 2 failed. A button that offers pour before boil ignores that plan. Easy and legal are different claims.

A plan, not a flat list.

1. Do 1, fill

2. Do 2, boil

3. Do 3, bag in cup

4. Do 4 only after 2 and 3

Pour is not legal yet

1. The 4 subtasks are fill, boil, bag in cup, and pour.

2. The plan allows fill and bag-in-cup to swap. It forbids pour before boil and pour before the bag.

3. A pour button shown before boil is easy and illegal. The analysis is allowed to say the easy action is the wrong action.

Problem 1. Which of the 4 tea subtasks may swap, and which ordering is forbidden?

Answer. Fill and bag-in-cup may swap. Pour cannot precede boil, and pour cannot precede the bag. A flat list of the 4 names hides those constraints. The plan is the analysis.

Problem 2. Step 2 failed because the water is missing. Is step 4 legal, and what does the plan say to do?

Answer. No. The rule is do 1 through 4, but if water is missing redo 1 before 4. A colleague can test the plan by asking exactly this. A pour button offered anyway ignores the plan.

Lecture 6 · 90 minutes

Styles are theories of action

An interaction style is a claim about how action is expressed. A command of 8 characters spends recall: the user must produce the string. A menu spends recognition: the options are on screen. Direct manipulation spends action on a visible object and shows the effect at once, and it should be reversible. A form spends action as values in labeled slots. You will see different errors from each. Recall fails by a blank memory. Recognition fails by a misleading label.

Selection rules in GOMS are how a skilled user picks a style when several methods exist. If the command is recalled, use the 2.35 second method. If it is not, use the menu. You can write that rule down. Interfaces get into trouble when the style and the task constraints conflict. A direct-manipulation canvas is a poor fit for entering 40

precise numbers. A form is a poor fit for rotating a shape by eye.

Recall against recognition.

Command

Produce 8 characters

Fails if memory is blank

Fast when recalled

Menu

Pick a visible option

Fails on a bad label

Slower in the model

Forty numbers, or one rotation

1. Entering 40 precise numbers wants labeled slots, a form. A canvas makes each number a hunting task.

2. Rotating a shape by eye wants direct manipulation: act on the object and see the effect at once, with undo.

3. If the 8-character command is recalled, the selection rule picks the 2.35 second method. If not, it picks the menu.

Problem 1. What error does a command style tend to produce, and what error does a menu tend to produce?

Answer. A command spends recall, so it fails by a blank memory, for example an 8-character string the user cannot produce. A menu spends recognition, so it fails by a misleading label. You will see different errors from each style.

Problem 2. Why is a direct-manipulation canvas a poor fit for 40 precise numbers, and a form a poor fit for rotating a shape by eye?

Answer. The canvas does not offer labeled slots for 40 exact values. The form does not let you act on a visible object and see a continuous rotation. Style has to match the task. The selection rule can still pick a recalled command at 2.35 seconds over the menu when the string is known.

Lecture 7 · 90 minutes

Where the thinking sits

Distributed cognition puts thinking across people and artifacts, not only inside one head. A wall chart that 4 people can see stores a memory none of them must recite. Move it to a personal screen and the audience may shrink from 4 people in the room to 1 login. The task 'who is next' can get worse on a sharper display. You ask where a fact lives, who updates it, and who can see the update.

A cognitive walkthrough reviews a step without a room full of users. You ask 4 questions. Will the person form this subgoal. Will they see the control. Will they connect the control to the subgoal. Will they notice progress. Failing 2 of the 4 on one screen is a result, not a hunch. If progress now sits on a display they are not looking at, question 4 fails. That is an evaluation gulf caused by moving the representation.

Four walkthrough questions.

Will they form the subgoal

Will they see the control

Will they connect it

Will they notice progress

The chart left the room

1. Four people could read the wall chart. The fact 'who is next' lived there.

2. The same fact moves to a personal screen. The audience becomes 1 login.

3. A walkthrough of the next step fails question 4 for the other 3 people: they cannot notice progress on a display they are not looking at.

Problem 1. A wall chart visible to 4 people moves to a personal screen. What changed about the audience, and what question do you ask?

Answer. The audience may shrink from 4 people in the room to 1 login. You ask where the fact lives, who updates it, and who can see the update. A sharper display can still make 'who is next' worse.

Problem 2. Progress is now on a display the person is not looking at. Which of the 4 walkthrough questions fails, and which gulf is that?

Answer. Question 4, will they notice progress. That is an evaluation gulf caused by moving the representation. Failing 2 of the 4 questions on one screen is a result of the review, not a hunch, and this review does not require a room full of users.

Lecture 8 · 90 minutes

An account, not a score

You can now give a theoretical account of an interface. Place a design against linked trails or against an editable constraint. Estimate a simple reaction at 240 milliseconds and do not add that figure to a keystroke sum. Name the larger gulf. Time a method with M, P, and K. Write a plan for 4 subtasks, including what may not be reordered. Say who can still see a fact if you move it off a shared wall.

The keystroke time and the gulf do not measure the same failure. The menu method at 3.95 seconds can still leave an evaluation gulf if the menu closes and the document looks unchanged. Your plan may say step 4 is illegal until step 2 finishes, while the screen offers step 4 anyway. You report both: the time estimate for the legal method, and the walkthrough failure on the illegal control.

Time and legality are different lines.

Keystroke time

Menu method is 3.95 seconds

Command is 2.35 if recalled

Plan and gulf

Step 4 is illegal early

No confirmation is a gulf

Fast, and still the wrong step

1. The legal menu method times at 3.95 seconds. The recalled command times at 2.35. You report the time of the legal method.

2. The plan says step 4 waits on step 2. The screen offers step 4 anyway. The walkthrough fails there.

3. You do not add the 3.95 and the walkthrough failure into one score. They are different mechanisms, and a single score would hide which one failed.

Problem 1. The menu method is 3.95 seconds and the document looks unchanged after the menu closes. The plan says step 4 is illegal until step 2 finishes, but the screen offers step 4. What do you report, and why not one combined score?

Answer. Report the 3.95 second estimate for the legal method, and report the walkthrough failure on the illegal control. Also name the evaluation gulf: the document looks unchanged, so the user cannot tell the goal was met. A single score would add those failures and hide the mechanism.

Problem 2. You estimated a simple reaction at 240 milliseconds and a command at 2.35 seconds. Why do those two numbers stay on separate lines?

Answer. The 240 millisecond figure is one perceptual, cognitive, and motor cycle. The 2.35 second figure is M + 5K. Adding them double-counts the motor act. You use each model for its own question.